Similarity

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Similarity

What are similar shapes?

- Two shapes are **similar** if they have the same **shape** and their **corresponding sides** are in **proportion**
  - One shape is an **enlargement** of the other
- **Similarity** does not imply **congruence**
  - Two similar shapes could be different enlargements of each other

How do we prove that two triangles are similar?

- To show that two **triangles** are **similar** you need to show that their **angles are the same**
  - If the angles are the same then **corresponding lengths** of a triangle will automatically be **in proportion**
- You can use angle properties to **identify equal angles**
  - Look out for for **isosceles triangles**, **vertically opposite angles** and **angles on parallel lines**
- If a question asks you to **prove two triangles** are **similar**
  - For **each pair** of corresponding angles
    - **State** that they are of **equal size**
    - Give a **reason** for why they are equal

How do we prove that two shapes are similar?

- To show that two **non-triangular shapes** are **similar** you need to show that their **corresponding sides** are **in proportion**
  - Divide the length of one side by the length of the corresponding side on the other shape to find the **scale factor**
- If the **scale factor** is the **same** for **all corresponding sides**, then the shapes are **similar**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

an hourglass formation. Look out for the **vertically opposite**, equal angles.

It may be helpful to sketch the triangles **next** to each other and facing in the **same direction**.

Worked Example

(a) Prove that the two rectangles shown in the diagram below are similar.

Answer:

Use the corresponding lengths (15 cm and 6 cm) to find the scale factor

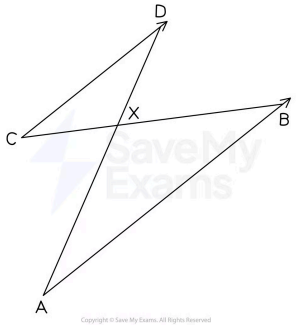
$$\frac{15}{6} = 2.5$$

Use the corresponding width (5 cm and 2 cm) to find the scale factor for the other pair of sides

$$\frac{5}{2} = 2.5$$

The two rectangles are similar, with a scale factor of 2.5

(b) In the diagram below, *AB* and *CD* are parallel lines. Show that triangles *ABX* and *CDX* are similar.



Answer:

State the equal angles by name, along with clear reasons
Don't forget to state that similar triangles need to have equal corresponding angles

Angle BAD = angle ADC (alternate angles on parallel lines are equal)

All three corresponding angles are equal, so the two triangles are similar

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